

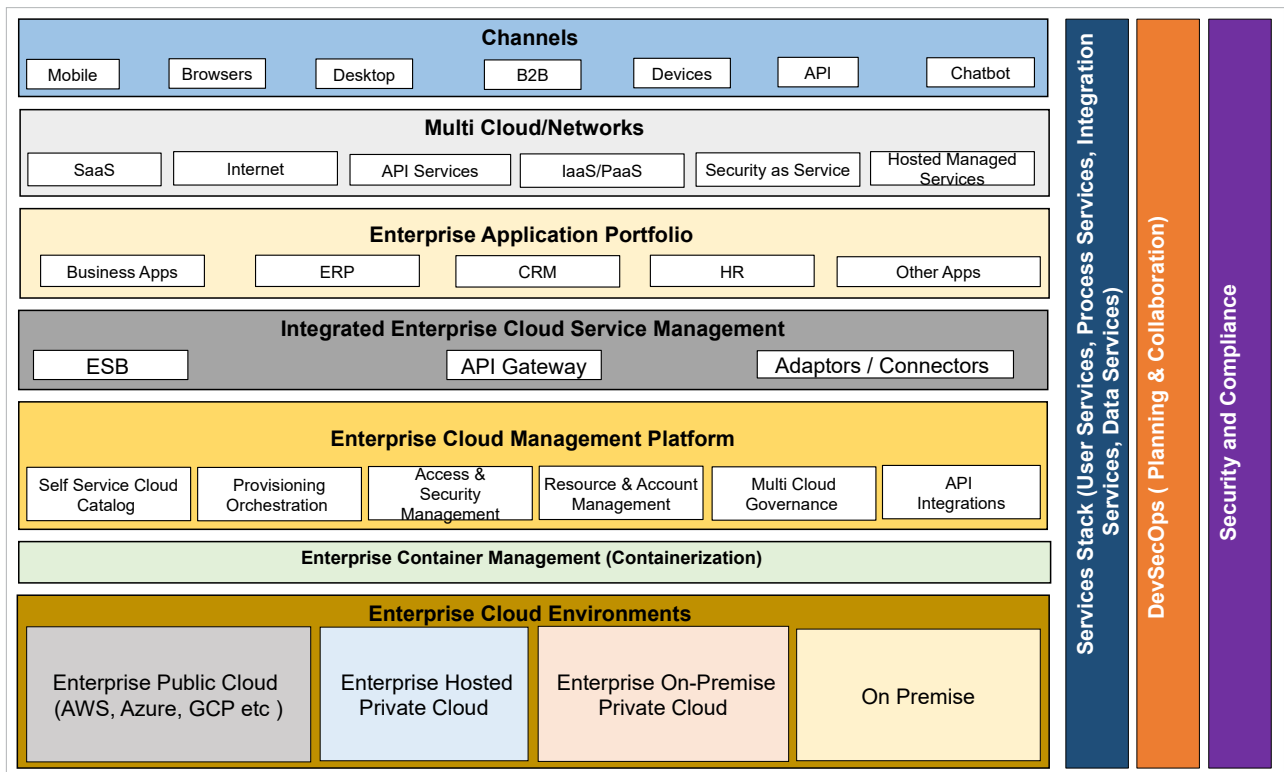


Figure 3: Hybrid cloud management model

Enterprise private cloud:

This refers to the hardware and software resources located within the enterprise's premises. Enterprise private cloud addresses the safety, security, regulatory and compliance requirements of the business. These resources are typically connected to the customer's on-premises/hosted environments through secure dedicated lines to allow secure and faster communication between applications.

On-premises private cloud:

This is used to host applications that need a high degree of co-location and controlled access to data and environments, and have regulatory and compliance requirements.

Legacy hardware: Applications requiring specialised hardware and point of delivery applications typically need to be hosted on-premises. They may not need scalability as the load is typically controlled, and if required is managed through vertical scaling of the environments.

SaaS platforms:

SaaS applications are typically consumed on a subscription basis. Here, the infrastructure and security are offered by the service provider.

Key open source cloud platforms

Open source cloud platforms can be chosen based on:

- Deployment flexibility
- Ease of operations
- Choice of application stacks
- Language, database and framework support
- Scaling capabilities
- Quality of service
- Tooling for development and operations
- Business fitment

Here are a few popular open source cloud platforms.

OpenStack: OpenStack is open source software for creating private and public clouds, built and disseminated by a large and democratic community of developers in collaboration with users.

It is mostly deployed as infrastructure-as-a-service (IaaS), whereby virtual servers and other resources are made available to customers. The software platform consists of interrelated components that control diverse, multi-vendor hardware pools of processing, storage, and networking resources throughout a data centre.

Cloud Foundry: Cloud Foundry is an open platform-as-a-service (PaaS) which provides a choice of clouds, developer frameworks and application services. The platform makes it faster and easier to build, test, deploy and scale applications.

OpenShift: This is Red Hat's cloud computing PaaS offering. It is an application platform in the cloud, where application developers and teams can build, test, deploy and run their applications.

Cloudify: Cloudify has been developed and designed on the principles of openness to power the IT transformation revolution. It enables